

Maximum Likelihood Estimation

Theory Questions with Solutions

TTK4260

Question 1 (*Probability vs Likelihood*)

Consider a Normal distribution with unknown mean μ and known variance $\sigma^2 = 1$. You observe a single data point $x = 2.3$.

- (a) Explain the key conceptual difference between *probability* and *likelihood*. What is fixed and what varies in each case? [4 points]
- (b) Write down the likelihood function $L(\mu \mid x = 2.3)$ for this observation. [3 points]
- (c) Does the likelihood function integrate to 1 over all possible values of μ ? Why or why not? [3 points]

Solution:

- (a) **Probability:** Parameters are **fixed** (known), and data is **variable** (unknown). We ask: “Given these parameters, what data might we observe?” The probability function integrates/sums to 1 over all possible data values.

Likelihood: Data is **fixed** (observed), and parameters are **variable** (unknown). We ask: “Given this observed data, which parameter values are most plausible?” The likelihood function does NOT integrate to 1 over the parameter space.

- (b) The likelihood function is:

$$L(\mu \mid x = 2.3) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2.3 - \mu)^2}{2}\right)$$

- (c) **No**, the likelihood function does not integrate to 1 over μ . The likelihood is not a probability distribution over parameters. In fact:

$$\int_{-\infty}^{\infty} L(\mu \mid x = 2.3) d\mu = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(2.3-\mu)^2/2} d\mu = 1$$

This is a coincidence for the Normal distribution; in general, $\int L(\theta) d\theta$ can be any positive number or even infinite.

Question 2 (*The Maximum Likelihood Principle*)

- (a) State the Maximum Likelihood Principle in words and write the formal definition of the MLE $\hat{\theta}_{\text{ML}}$. [4 points]
- (b) Compare Maximum Likelihood with Least Squares. List two things that ML provides that LS does not. [4 points]
- (c) What is the main caveat or disadvantage of using Maximum Likelihood compared to Least Squares? [2 points]

Solution:

- (a) **In words:** The Maximum Likelihood Principle states that we should choose the parameter value that makes the observed data most probable.

Formal definition:

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta \in \Theta} L(\theta \mid \mathcal{D}) = \arg \max_{\theta \in \Theta} P(\mathcal{D} \mid \theta)$$

- (b) ML provides things that LS does not:

- **Uncertainty quantification:** ML enables confidence intervals and hypothesis tests through the Fisher Information framework.
- **Probabilistic interpretation:** ML provides a complete probabilistic model, allowing us to compute probabilities of future observations.

- (c) The main caveat is that **ML requires distributional assumptions**. If the assumed distribution is wrong (model misspecification), the MLE may be biased or inefficient. LS makes no distributional assumptions, so it is more robust in this sense.

Question 3 (*Why Log-Likelihood?*)

For i.i.d. observations $x_1, x_2, \dots, x_n$, the likelihood function is $L(\theta) = \prod_{i=1}^n f(x_i \mid \theta)$.

- (a) Explain why computing $L(\theta)$ directly can cause numerical problems when n is large. Give a concrete numerical example. [4 points]
- (b) Define the log-likelihood $\ell(\theta)$ and explain why $\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta)$. [3 points]
- (c) List two additional advantages of working with log-likelihood beyond numerical stability. [3 points]

Solution:

- (a) The likelihood is a product of n probabilities, each typically less than 1. For large n , this product becomes extremely small, causing **numerical underflow** (the computer represents it as 0).

Example: If each $f(x_i \mid \theta) \approx 0.1$ and $n = 1000$:

$$L(\theta) = 0.1^{1000} = 10^{-1000}$$

This is far smaller than the smallest representable floating-point number ($\approx 10^{-308}$), so it underflows to 0.

- (b) The log-likelihood is defined as:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i \mid \theta)$$

Since log is a **monotonically increasing** function, the value of θ that maximizes $L(\theta)$ is the same as the value that maximizes $\log L(\theta)$:

$$\arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta)$$

- (c) Two additional advantages:

- **Analytical convenience:** Products become sums ($\log(ab) = \log a + \log b$), and exponentials simplify ($\log(e^x) = x$). Derivatives of sums are much easier to compute than derivatives of products.
- **Optimization properties:** The log-likelihood is often concave, making it easier to find the global maximum using standard optimization methods like gradient descent.

Question 4 (*The Score Function*)

The score function is defined as $s(\theta) = \frac{\partial}{\partial \theta} \ell(\theta)$.

- What equation must be satisfied at the MLE? State this in terms of the score function. [2 points]
- What second-order condition should you check to confirm that a solution to the score equation is actually a maximum (not a minimum)? [3 points]
- For the Poisson distribution with PMF $P(X = k \mid \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$, write down the log-likelihood for a single observation x . [3 points]

Solution:

- At the MLE, the score function equals zero:

$$s(\hat{\theta}_{\text{ML}}) = 0 \quad \text{or equivalently} \quad \left. \frac{\partial \ell}{\partial \theta} \right|_{\theta=\hat{\theta}_{\text{ML}}} = 0$$

- To confirm a maximum (not minimum), check that the **second derivative is negative**:

$$\left. \frac{\partial^2 \ell}{\partial \theta^2} \right|_{\theta=\hat{\theta}_{\text{ML}}} < 0$$

For multiple parameters, the Hessian matrix $\nabla^2 \ell(\hat{\theta})$ should be **negative definite**.

- For Poisson with PMF $P(X = x \mid \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$:

$$\ell(\lambda) = \log \left(\frac{\lambda^x e^{-\lambda}}{x!} \right) = x \log \lambda - \lambda - \log(x!)$$

Question 5 (*MLE for Normal Distribution: Derivation*)

Let $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown.

- Write down the likelihood function $L(\mu, \sigma^2)$ for these observations. [3 points]
- Derive the log-likelihood $\ell(\mu, \sigma^2)$ and simplify it. [4 points]
- Find $\hat{\mu}_{\text{ML}}$ by taking the derivative of ℓ with respect to μ , setting it to zero, and solving. [4 points]
- Find $\hat{\sigma}_{\text{ML}}^2$ by taking the derivative of ℓ with respect to σ^2 , setting it to zero, and solving. [4 points]

Solution:

(a) The likelihood function is:

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

(b) Taking the log:

$$\begin{aligned}\ell(\mu, \sigma^2) &= \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right] \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\end{aligned}$$

(c) Taking derivative with respect to μ :

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

Setting to zero:

$$\sum_{i=1}^n (x_i - \mu) = 0 \implies \sum_{i=1}^n x_i = n\mu \implies \boxed{\hat{\mu}_{\text{ML}} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i}$$

(d) Taking derivative with respect to σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2$$

Setting to zero and solving:

$$\frac{n}{2\sigma^2} = \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 \implies \boxed{\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Question 6 (MLE for Bernoulli Distribution)

A coin is flipped n times, yielding outcomes $X_1, X_2, \dots, X_n$ where $X_i \in \{0, 1\}$ and $P(X_i = 1) = p$.

(a) Write down the likelihood function $L(p)$ for these observations. Simplify it in terms of $\sum_{i=1}^n x_i$. [4 points]

(b) Derive the MLE $\hat{p}_{\text{ML}}$ by maximizing the log-likelihood. Show your work. [5 points]

Solution:

(a) The likelihood function is:

$$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i} = p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i}$$

(b) Log-likelihood:

$$\ell(p) = \left(\sum_{i=1}^n x_i \right) \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

Taking derivative:

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p}$$

Setting to zero:

$$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1-p} \implies (1-p) \sum x_i = p(n - \sum x_i)$$

$$\sum x_i - p \sum x_i = pn - p \sum x_i \implies \sum x_i = pn$$

$$\boxed{\hat{p}_{\text{ML}} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}}$$

Question 7 (*MLE for Poisson Distribution*)

A call center records the number of calls received per hour. The data follows a Poisson distribution with unknown rate λ .

- (a) Write down the likelihood function for n observations $x_1, \dots, x_n$. [3 points]
- (b) Derive the log-likelihood and find the MLE $\hat{\lambda}_{\text{ML}}$. [5 points]
- (c) Given data: 4, 7, 5, 6, 8, 3, 5, 6, 7, 4 calls per hour over 10 hours, compute $\hat{\lambda}_{\text{ML}}$. [2 points]
- (d) For the Poisson distribution, what is the relationship between the mean and variance? How can you use this to check whether Poisson is an appropriate model? [3 points]

Solution:

(a) The likelihood function is:

$$L(\lambda) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod_{i=1}^n x_i!}$$

(b) Log-likelihood:

$$\ell(\lambda) = \left(\sum_{i=1}^n x_i \right) \log \lambda - n\lambda - \sum_{i=1}^n \log(x_i!)$$

Taking derivative:

$$\frac{d\ell}{d\lambda} = \frac{\sum x_i}{\lambda} - n$$

Setting to zero:

$$\frac{\sum x_i}{\lambda} = n \implies \boxed{\hat{\lambda}_{\text{ML}} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}}$$

(c) Sum = 4 + 7 + 5 + 6 + 8 + 3 + 5 + 6 + 7 + 4 = 55

$$\hat{\lambda}_{\text{ML}} = \frac{55}{10} = 5.5 \text{ calls per hour}$$

- (d) To check if Poisson is appropriate, compare the sample mean $\bar{x}$ with the sample variance s^2 . If they are approximately equal ($s^2 \approx \bar{x}$), Poisson is reasonable. If $s^2 \gg \bar{x}$ (overdispersion), consider Negative Binomial instead.

Question 8 (*MLE for Exponential Distribution*)

Component lifetimes follow an Exponential distribution with rate parameter λ , i.e., $f(x | \lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$.

- (a) Write down the likelihood function for n observed lifetimes $x_1, \dots, x_n$. [3 points]
- (b) Derive the MLE $\hat{\lambda}_{\text{ML}}$. [5 points]

Solution:

- (a) The likelihood function is:

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

- (b) Log-likelihood:

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

Taking derivative:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i$$

Setting to zero:

$$\frac{n}{\lambda} = \sum_{i=1}^n x_i \implies \boxed{\hat{\lambda}_{\text{ML}} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}}$$

Question 9 (*ML = LS Connection*)

Consider the linear regression model $y_i = \mathbf{x}_i^\top \boldsymbol{\theta} + \varepsilon_i$ where $\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$.

- (a) Write down the likelihood function $L(\boldsymbol{\theta}, \sigma^2 | \mathbf{y}, \mathbf{X})$ for this model. [4 points]
- (b) Show that maximizing the log-likelihood with respect to $\boldsymbol{\theta}$ is equivalent to minimizing $\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2$. [5 points]
- (c) Under what assumption about the error distribution does ML = LS? [2 points]

Solution:

- (a) Since $y_i | \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\theta}, \sigma^2)$:

$$L(\boldsymbol{\theta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2}{2\sigma^2}\right)$$

(b) Log-likelihood:

$$\ell(\boldsymbol{\theta}, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2$$

To maximize w.r.t. $\boldsymbol{\theta}$, note that the first term doesn't depend on $\boldsymbol{\theta}$. We maximize:

$$-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2$$

Since $\sigma^2 > 0$, this is equivalent to **minimizing**:

$$\sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\theta})^2$$

which is exactly the **Least Squares objective**. $\square$

(c) ML = LS when errors are **Gaussian (Normal)**.

Question 10 (*Choosing the Right Distribution*)

(a) For each scenario, name the most appropriate distribution:

- Number of customer arrivals per hour at a store
- Whether a manufactured part passes or fails quality inspection
- Time until a light bulb burns out
- Heights of adult males in a population

[4 points]

(b) Explain why using a Normal distribution for count data (e.g., number of defects) would be problematic.

[3 points]

(c) You have count data where the sample variance is much larger than the sample mean. What does this suggest, and what distribution might be more appropriate than Poisson?

[3 points]

Solution:

- (a)
- Customer arrivals per hour: **Poisson** (counts in fixed time, independent arrivals)
 - Pass/fail inspection: **Bernoulli** (binary outcome)
 - Time until light bulb burns out: **Exponential** (waiting time with constant hazard rate)
 - Heights of adult males: **Normal** (continuous, symmetric, sum of many small effects)

(b) Problems with Normal for count data:

- Normal is **continuous** while counts are **discrete integers**
- Normal can take **negative values**, but counts are always ≥ 0
- Normal assumes symmetric errors, but count data is often right-skewed

(c) If sample variance $\gg$ sample mean, this indicates **overdispersion**. Poisson requires $\text{Var}(X) = \text{Mean}(X) = \lambda$.

A more appropriate distribution is the **Negative Binomial**, which has an extra parameter allowing $\text{Var}(X) > \text{Mean}(X)$.